Tiny-Dataset Scale-Consistent Diffusion Framework for Single SAR Image Building Height Estimation

Qing Li, Yulun Wu, *Member, IEEE*, Heng Zhang, *Member, IEEE*, Jili Wang, *Member, IEEE*, Fengli Xue, *Member, IEEE*, Zhen Chen, *Member, IEEE*, and Fengjun Zhao

***Abstract*—Recent learning-based methods for building height estimation from a single SAR image have shown promising performance. However, existing methods are constrained by limited model learning capacity and their dependence on large-scale SAR-LiDAR paired datasets. To address these issues, this article proposes a tiny-dataset scale-consistent diffusion framework, which comprises the following three components: 1) a stable diffusion-based model is fine-tuned using just one GaoFen-3 (GF-3) SAR image to achieve task transfer. Dedicated data augmentation strategies are introduced and validated through ablation studies; 2) a novel and fast height ground truth pipeline is designed to generate radar-domain building facade points from sparse LiDAR data, with vegetation effects explicitly removed. Experimental analysis reveals that removing vegetation leads to improved building height estimation accuracy; and 3) a geometry-driven transformation module is introduced to ensure a consistent height scale, making it universally applicable across images. Experiments on GF-3 spotlight SAR data demonstrate that the proposed framework, trained on only one image, outperforms current UNet- and Transformer-based baselines. Cross-sensor evaluation on TerraSAR-X further indicates that the proposed framework has a certain degree of generalization ability.**

***Index Terms*—Diffusion model, GaoFen-3 (GF-3), scale consistency, single SAR image height estimation, task transfer, tiny dataset.**

I. Introduction

BUILDINGS are key elements of urban landscapes, and their height information is vital for applications such as population density estimation [1] and urban environmental analysis [2], [3], [4], [5]. Traditional 3-D reconstruction relies on multiview data: LiDAR achieves the highest accuracy but at high cost, while optical methods (stereo or oblique photogrammetry) and SAR-based methods (InSAR, TomoSAR) require image pairs or stacks from different viewing angles. In contrast, estimating building heights from a single satellite-pass SAR image is highly appealing. SAR is insensitive to cloud cover and weather, enabling fast and near-real-time height estimation, which is particularly valuable for disaster monitoring and emergency response.

However, single-image SAR-based building height estimation presents distinctive challenges. Recovering height from a single observation is inherently ill-posed. Unlike optical imagery, SAR employs a side-looking imaging geometry, so buildings are affected by distortions such as layover and shadow, which complicate the relationship between image appearance and physical height [6], [7]. In addition, speckle noise and complex scattering mechanisms make urban structures harder to interpret [8], [9]. Reliable supervision is also difficult to obtain, as high-quality radar-domain height labels typically require precise co-registration with auxiliary data such as LiDAR. These factors make a single-image SAR height estimation fundamentally different from a direct transfer of conventional monocular optical depth estimation methods.

In remote sensing, research on single-image height estimation has been primarily developed in the optical domain. Among the pioneering works, Mou and Zhu [10] first introduced convolutional neural networks (CNNs) for pixel-wise height estimation from monocular optical imagery in 2018. Subsequent work further improved performance through enhanced preprocessing, multiscale feature extraction, progressive learning strategies, Transformer-based architectures, and recent large-scale applications [11], [12], [13], [14], [15]. Another line of work explored mappings to normalized digital surface models (nDSMs) [16], [17]. Together, these studies collectively demonstrate the potential of deep learning for single-image height estimation. However, this line of research still relies largely on supervised architectures and annotated training data. Such conditions are difficult to satisfy in SAR-based building height estimation, where high-quality labels are much harder to obtain.

More broadly, height estimation from a single SAR image can draw inspiration from the rapid advances in monocular optical image depth estimation. Model capacity has steadily increased, from relatively modest CNNs to large transformer-based architectures [18], [19]. More recently, diffusion models have received increasing attention because large-scale pretraining provides strong visual priors [20]. Several methods have employed diffusion models for depth estimation [21], [22], [23], [24]. Marigold-like methods further showed that

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pretrained latent diffusion models (LDMs) can be used for image-conditioned depth estimation [25], [26], and recent improvements in inference efficiency have made fine-tuning for downstream tasks more feasible [27]. These results indicate that pretrained diffusion models may also be useful for single-image height estimation under limited supervision. However, current evidence is still mainly limited to optical imagery, and its applicability to SAR-based building height estimation remains to be verified.

In recent years, research on urban height estimation from single SAR images has begun to achieve notable progress. Existing studies can be broadly grouped into two directions. One direction focuses on dense height prediction. The first deep learning-based approach for single SAR image height prediction was introduced by Recla and Schmitt [28]. It demonstrated the feasibility of predicting urban height from very high-resolution (VHR) TerraSAR-X images. Recla and Schmitt [29] further proposed the SAR2Height framework in 2024. By incorporating sensor knowledge into the UNet-based model, their method enabled fast height estimation and was validated through city-wise cross-validation on more than 50 TerraSAR-X images from eight cities. Subsequently, they extended their work to change detection [30], as well as multitrack height reconstruction and fusion [31], [32]. This evolution also reflects a shift from early feasibility studies toward training settings supported by larger multicity datasets. The other direction considers building-level height estimation. In 2022, Sun et al. [7] introduced GIS-derived building footprints as auxiliary information, enabling rapid building-wise height estimation. More recently, object-based deep learning methods further extended this direction through multicity training and large-scale sample construction [33]. These studies confirm the feasibility of single-image SAR building height estimation, while also indicating that substantial data preparation is still often required.

Recent studies have also started to introduce diffusion models into SAR-related tasks, particularly in work published from 2024 to 2025, including SAR-to-optical translation [34], [35], target detection [36], [37], despeckling [38], and image generation [39]. However, these studies still mainly focus on the tasks listed above and have not yet clearly established their use in single-image SAR building height estimation, especially for marigold-like diffusion frameworks [25], [26].

Despite these achievements, single-image SAR building height estimation still faces several practical limitations. First, most existing methods remain mainly supervised and have not yet fully exploited the priors available in large pretrained generative models. Second, dataset preparation remains costly, as current methods typically depend on VHR SAR images with accurately aligned LiDAR or related height references. Third, the geometric mismatch between physical height and SAR image representation further limits model transferability.

To address these challenges, this work proposes a tiny-dataset scale-consistent diffusion framework for single-image SAR building height estimation (see Fig. 1). The framework is designed for SAR-domain adaptation under extremely limited supervision. The main contributions of this work are summarized as follows.

1) We propose to fine-tune a stable diffusion-based model using a single GaoFen-3 (GF-3) SAR image. It leverages prior knowledge from optical geometry and achieves task transfer to the SAR domain. We design dedicated data augmentation strategies for training on a tiny dataset.
2) We develop a ground-truth generation pipeline to generate vegetation-free radar-domain building facades from sparse LiDAR points.
3) We propose a geometry-driven transformation module that converts physical building heights into a consistent scale space, enabling the model to be universally applicable.
4) We evaluate the proposed framework on GF-3 Spotlight (SL) SAR images from unseen regions and TerraSAR-X data, showing superior performance over UNet- and Transformer-based baselines and a certain degree of generalization ability.

The remainder of this article is organized as follows. Section II describes the scale-consistent data preparation pipeline for training and testing, which consists of the first three components in the framework shown in Fig. 1. Section III presents the remaining model-related steps, detailing the theory and implementation for fine-tuning the diffusion model. Section IV describes the experimental data used, the experimental results and analysis, as well as the ablation studies. Section V provides further discussion, and Section VI offers conclusions.

II. Scale-Consistent Data Preparation Pipeline for Training and Testing

As the model is trained in a supervised manner, this chapter begins with the creation of a dataset. The first two sections describe the preprocessing of the training SAR data and the generation of corresponding radar-domain ground truth. The proposed ground-truth generation process addresses the issue of missing points on vertical facades in LiDAR data. To resolve the inconsistency of building height scales across SAR images with varying resolutions and incidence angles, the second section further introduces a geometry-driven method to enforce scale consistency of height values. Subsequently, the third section presents the processing of test images, including both preprocessing and postprocessing procedures. Finally, data augmentation strategies specifically designed for tiny datasets are introduced. In summary, this chapter covers the three remaining components of the framework illustrated in Fig. 1, excluding the model-related part.

A. Preprocessing of Training SAR Data

During training, SAR amplitude information is typically used as the input. However, SAR data acquired under different imaging geometries exhibit amplitude inconsistencies. Therefore, it is necessary to perform radiometric correction using the metadata file to convert the amplitude information into the backscattering coefficient σ_{dB}. The formula for calculating σ_{dB} of the GF-3 Level-1A product used is as follows:

$$\sigma_{\mathrm{dB}} = 10\lg\left[P\left(\mathrm{QualifyValue}/32767\right)^2\right] - K_{\mathrm{dB}} \quad (1)$$

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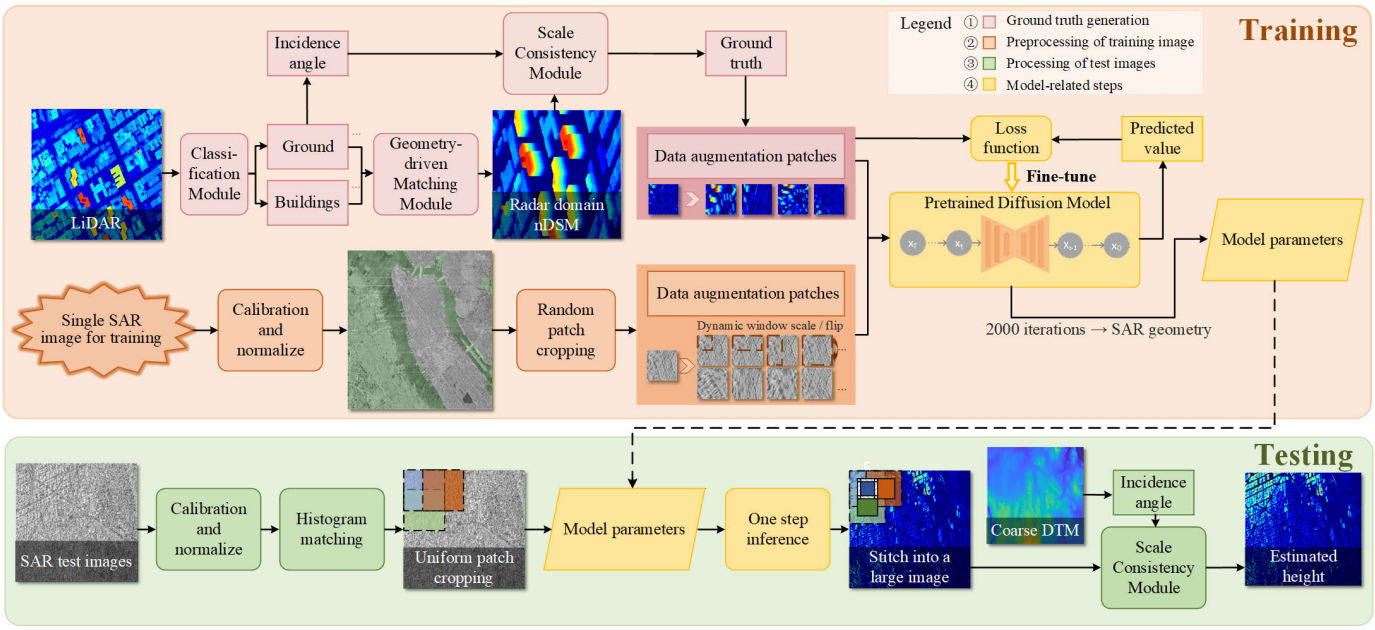


Fig. 1. Overview of the tiny-dataset scale-consistent diffusion framework. Four colors indicate its four components: ground truth generation, preprocessing of training images, processing of test images, and model-related steps.

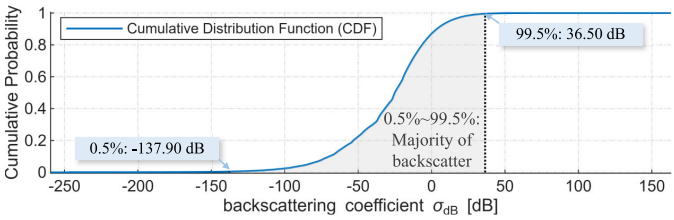


Fig. 2. Cumulative distribution function (cdf) of the training SAR image backscattering coefficient σ_{dB}.

where $P = I^2 + Q^2$, I is the real part of the image, and Q is the imaginary part of the image. QualifyValue is the image quantization value, and K_{dB} is the radiometric calibration constant. These two parameters are provided by the product description file.

The cumulative probability distribution of the σ_{dB} from the training SAR image is shown in Fig. 2. The dB values at the lower and upper 0.5% quantiles are obtained. A small margin is retained, and the valid range is finally clipped to $[-140, 45]$ dB before being normalized to $[0, 1]$.

In contrast to SAR2Height [29], which leverages more than 50 TerraSAR-X VHR images acquired from eight different cities, our experiment is designed to explore the challenge of small-dataset training using data from a single SAR image. To this end, we selected a high-quality GF-3 SL SAR image over New York City (NYC) as the experimental sample. The image covers an area of approximately 10×8 km, encompassing both high-rise and low-rise buildings with complex layover and shadowing effects.

Considering that the skyscraper area of NYC contains multiple buildings exceeding 400 m in height, with the tallest one surpassing 500 m, a single high-rise can extend across more than 600 pixels in the range direction of the selected SAR image. To expand the model's receptive field and mitigate the truncation of high-rise buildings during training and testing, multilook processing is essential for SAR image preprocessing. In this study, the double multilook result of the original SAR image was adopted as the standard input image, and the annotation of height ground truth was also performed based on this double multilook SAR image. The test samples were sampled to approximately the same spatial resolution for fair evaluation. Following the patch-size settings in several reference studies [25], [26], and considering both empirical performance and GPU memory constraints, the training patch size was set to 640×640 pixels, which can be effectively trained on an NVIDIA V100S GPU.

Furthermore, due to the limited coverage of LiDAR-derived ground-truth heights, regions without reliable height measurements, such as areas over water bodies and construction zones with inaccurate LiDAR data, were masked out. The pixel size of the SAR image after double multilook processing is [14 987, 4891], and the number of pixels in the valid region after masking is 47 162 565, accounting for 64.3% of the total pixels of the image.

B. Scale-Consistent Ground Truth Generation

To obtain height truth labels registered with SAR images, it is necessary to project LiDAR height values (typically in UTM projection) into the radar slant range domain. The objective is to generate the target output for the model—the height truth of buildings in the SAR image domain. The general idea is to geocode the precise digital surface model (DSM) and digital terrain model (DTM) into the slant-range domain, respectively, and then subtract the corresponding DTM height from the DSM height to obtain the nDSM. However, numerous challenges arise in practical processing. Therefore, this study designs a workflow for ground truth generation and ultimately

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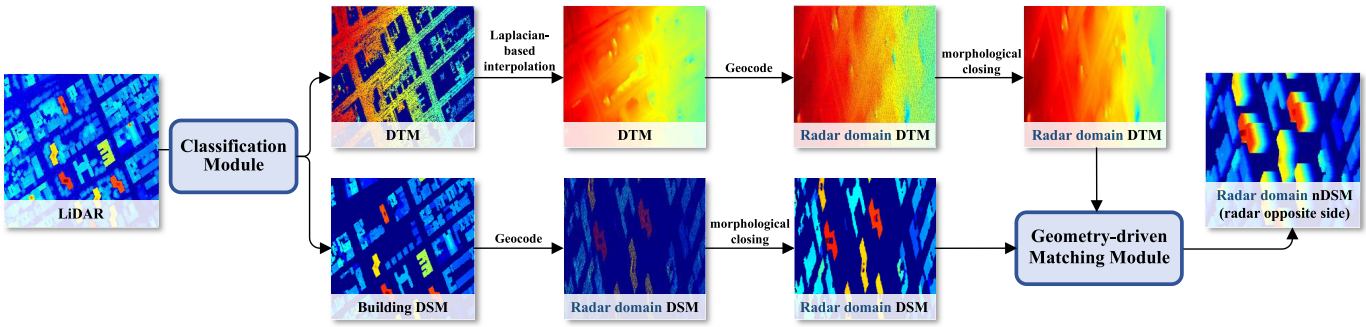


Fig. 3. Workflow for generating the nDSM ground truth.

converts the ground truth into scale-consistent, dimensionless input values for the model. The detailed procedures are described in Section II-B1–II-B3.

1) Principle of Geocoding: It locates each point in the LiDAR data within the SAR image domain using the Range-Doppler equations, which are defined by the intersection of three fundamental relationships [40].

The Earth's shape model serves as an inertial reference frame with respect to the Earth's center of mass. Typically, an oblate ellipsoid is used to represent this model

$$\frac{x_t^2 + y_t^2}{(R_e + h)^2} + \frac{z_t^2}{R_p^2} = 1 \tag{2}$$

where R_e is the mean equatorial radius, and h represents the absolute height relative to the reference ellipsoid. $R_p = (1 - f)(R_e + h)$ is the polar radius, where f is the flattening factor. x_t, y_t, and z_t are the three components of target position vectors $\mathbf{R_t}$ in this reference frame.

The second equation is the Doppler equation, which defines the central plane of the radar beam at a specific moment in time

$$f_D = \frac{2}{\lambda R}(\mathbf{V_s} - \mathbf{V_t}) \cdot (\mathbf{R_s} - \mathbf{R_t}) \tag{3}$$

where $\cdot$ denotes the vector dot product, f_D is the Doppler central frequency, λ is the wavelength, and R refers to the slant range in SAR images. $\mathbf{V_s}$ and $\mathbf{V_t}$ are the sensor's and the target's velocity, and $\mathbf{V_t}$ is determined by $\mathbf{V_t} = \boldsymbol{\omega_e} \times \mathbf{R_t}$ with $\boldsymbol{\omega_e}$ acting as Earth's rotational velocity vector. $\mathbf{R_s}$ is the sensor's position.

Within this plane, the range equation is finally used to locate the desired point

$$R = [(\mathbf{R_s} - \mathbf{R_t}) \cdot (\mathbf{R_s} - \mathbf{R_t})]^{1/2}. \tag{4}$$

The range R is determined by the positions of the sensor and the target point, and it is computed as the square root of the dot product of their vector difference.

By simultaneously solving the three equations, a point in the LiDAR geodetic coordinate system can be directly mapped to the SAR image domain.

However, several practical challenges arise during data processing. First, in many LiDAR datasets, building facade heights are often sparse or even missing, whereas facades dominate the SAR image domain. Second, after projection into the radar domain, the points cannot fully cover all azimuth and range positions, resulting in numerous gaps. Layover makes it infeasible to fill the gaps through simple interpolation. Moreover, obtaining accurate DTM products in complex urban areas is costly.

2) SAR-Domain Height Ground-Truth Generation Workflow: This study proposes a processing workflow in which the LiDAR point cloud is first classified into building and ground points and then geocoded separately into the radar domain. Subsequently, the nDSM is generated through a geometry-driven matching module. The overall workflow is illustrated in Fig. 3, and the detailed procedure is described below.

a) Classification module: The primary advantage of classifying the point clouds is that it enables convenient, separate processing of ground and rooftop points. In most cases, only rooftop points experience partial layover. Second, the use of SAR imaging geometry enables the generation of complete building facades by matching rooftop points with the corresponding ground points at building footprints. Third, as the ground surface is generally smooth, an accurate DTM can be produced through simple interpolation.

In practice, the point cloud classification can be performed using the LAStools software. After classification, the points labeled as "building" are rasterized using a maximum-value rule to obtain a DSM that retains only rooftop elevations. The points labeled as "ground" are rasterized using a minimum-value rule to produce a DTM containing voids. These voids are subsequently filled within building footprints using Laplacian-based interpolation to create a complete DTM.

In the current implementation, the "building DSM" and "DTM" in Fig. 3 are generated from the points labeled as "building" and "ground," respectively, while points from other classes, such as vegetation, are not used. The resulting ground truth, therefore, represents vegetation-free building heights. If all above-ground objects are to be included, the "building DSM" can be replaced by an all-object DSM. The latter is generated by rasterizing all LiDAR points with a maximum-value rule and comparing the result pixel-wise with the ground-derived DTM. Pixels with missing DTM values or height differences above a predefined threshold (e.g., 2 m) are retained.

b) Geocode and morphological closing: Both the DSM and DTM are then geocoded into the SAR image domain. Since rooftop regions of different buildings may overlap in radar geometry, a nearest-neighbor interpolation method with

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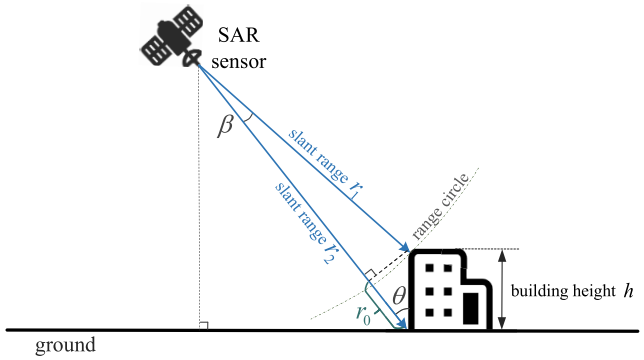


Fig. 4. Schematic of spaceborne SAR image geometry (not to scale for illustrative purposes).

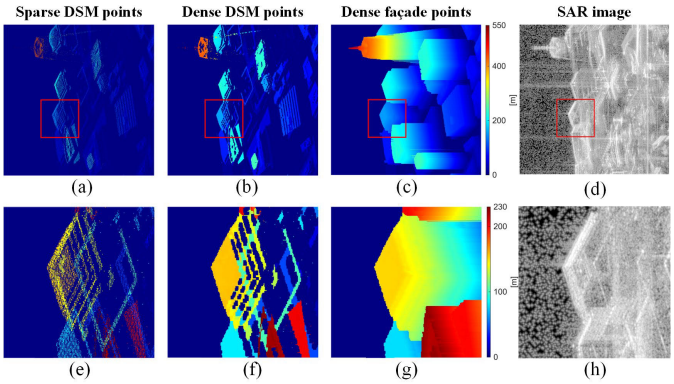


Fig. 5. Radar-domain building facade generation. The example is taken from a 1000 × 1000 pixel region in the NYC training set. (a) Building DSM geocoded to the radar domain. (b) Result of applying morphological closing to (a). (c) Facade generated by the matching module based on (b). (d) SAR image of the corresponding area. (e)–(h) Enlarged views of the red-boxed regions in (a)–(d), respectively.

a maximum search radius of one pixel is adopted during geocoding to prevent unrealistic height values (e.g., values between tall and short buildings) that may arise from other interpolation schemes. The same nearest-neighbor interpolation is also applied to the DTM, which rarely suffers from layover effects.

Discontinuities and voids resulting from geocoding can be mitigated using appropriate filtering operations. Following [29], in overlapping regions, the maximum value is taken as the valid height. Therefore, a morphological closing operation is applied to the rooftop DSM to fill voids and replace lower elevations in overlapping regions, ensuring that the highest rooftop values are retained. In the experiment, two successive closing operations with sliding windows of size 3 × 3 and 5 × 5 were performed, each implemented by a maximum filter followed by a minimum filter. After morphological closing, higher values in overlapping areas are preserved and voids are filled. Higher values in overlapping areas are preserved and voids are filled. For the DTM, voids are filled via repeated mean filtering until all gaps are closed. All filtering operations are implemented using the ndimage module of the SciPy library. As a result, complete DSM and DTM datasets geocoded into the radar geometry are obtained.

c) Geometry-driven matching module: The DTM and DSM in the radar domain are matched based on the SAR imaging geometry. Since our goal is to obtain nDSM, nonbuilding areas can simply be set to zero. Therefore, it is only necessary to match each roof point of the building with the ground value at its corresponding footprint location. The theoretical basis for this matching lies in the SAR imaging geometry: along the range direction, the heights of building facade points decrease with increasing range, and the rate of decrease depends on the incidence angle and range resolution at each point.

To facilitate understanding, the geometric relationship between the height difference and the slant range difference of a roof point and its corresponding ground point is first analyzed. The geometry of the SAR image is shown in Fig. 4, where the building height is h, the slant range from the SAR satellite to the top of the building is r_1, the slant range to the bottom of the building is r_2, the incident angle in the building area (defined as the angle between the ellipsoid normal and the look vector) is θ, and the angle between the SAR echoes corresponding to the two slant ranges is β. For spaceborne SAR satellites, the slant ranges r_1 and r_2 are much larger than the building height h, so β is very close to 0°. In the figure, it can be approximately considered that

$$r_2 = r_1 \cdot \cos\beta + r_0 \approx r_1 + r_0. \tag{5}$$

Therefore, the building height h and the slant range difference $(r_2 - r_1)$ have the following relationship:

$$h = \frac{r_0}{\cos(\theta)} \approx \frac{r_2 - r_1}{\cos(\theta)}. \tag{6}$$

Similarly, for two adjacent pixels located on a building facade in the slant range domain, their slant-range difference equals the range pixel spacing ρ_r, which is provided in the SAR parameter file. The incidence angle of points on building facades is estimated as the incidence angle at the corresponding footprint points. With this approach, the local incidence angle matrix $\boldsymbol{\Theta}$ for the entire SAR image can be approximately computed using the DTM in the slant range domain together with the SAR orbital parameter file. In practice, this can be implemented with the gc_map2 command in the Gamma software package. Then, analogous to (6), the height along the slant range direction gradually decreases, and the absolute value of the height gradient is defined as the **height pixel spacing**. Accordingly, the height pixel spacing matrix $\boldsymbol{\rho}_h$ corresponding to each pixel in the SAR image is expressed as

$$\boldsymbol{\rho}_{h,ij} = \frac{\rho_r}{\cos\left(\boldsymbol{\Theta}_{ij}\right)} \tag{7}$$

where the subscript ij represents the (i, j)th element of the corresponding matrix.

Next, we introduce an algorithm that utilizes height pixel spacing to match building points with their corresponding ground points. For each building point, a 1-D search is conducted along the range direction to identify the corresponding ground footprint, as detailed in Algorithm 1.

In Algorithm 1, the resulting $\mathbf{h}_{facade}$ represents the nDSM in the RD domain, while $\mathbf{p}_{count}$ corresponds to the pixel count, which is described in detail in Section II-B3. The facade

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Algorithm 1 1-D Matching of Rooftop Points to Ground Footprint Points

Input: n: total number of range pixels. h_{roof}: rooftop height at the ith range pixel. $\mathbf{h}_{DTM}, \boldsymbol{\rho}_{range} \in \mathbb{R}^n$: DTM height and height pixel spacing along the range line.
Output: $\mathbf{h}_{facade}$ the facade height vector and $\mathbf{p}_{count}$ the pixel count vector (will be detailed in Sec. II-B3) along the range line.
Step 1: Search for the best-matched ground point.
for $j = i + 1$ to n **do**
$\hat{h}_{DTM} = h_{roof} - \boldsymbol{\rho}_{range}[j] \cdot (j - i)$
$e = \hat{h}_{DTM} - \mathbf{h}_{DTM}[j]$
$\mathbf{e}_{\text{vec}} \leftarrow [\,\mathbf{e}_{\text{vec}},\, e\,]$
end for
$\min_e = \min_i |e_i|$
$g = \arg\min_i |e_i|$
Step 2: Generate the facade vector.
if $\min_e < Tol$:
$\mathbf{p}_{count}[i:g] = [g - i - k \text{ for } k \text{ in } 1 \text{ to } g - i]$
$\mathbf{h}_{facade}[i:g] = \boldsymbol{\rho}_{range}[g] \cdot \mathbf{p}_{count}[i:g]$

generation results for a 1000 × 1000 pixel region in the NYC training set are shown in Fig. 5. The tolerance parameter Tol is associated with the geometric discretization of the matching process. Sensitivity analysis shows that the generated ground truth becomes stable when $Tol \geq 3$ m. Therefore, $Tol = 3$ m is adopted as the smallest value within the stable regime. Detailed statistical analysis and supporting evidence are provided in the supplementary material.

3) Scale Consistency Module: Due to variations in range resolution and incidence angle, the scale of height ground truth differs across SAR images, hindering unified training and testing. To overcome this problem, we design a scale consistency module (SCM). Although the physical heights may differ, the imaging geometry of SAR provides a natural and consistent scale that links height to image space, namely, the number of pixels a building spans along the range direction. By dividing the physical height by the height pixel spacing in (7), the corresponding pixel count can be conveniently obtained. Similarly, the incidence angle matrix $\boldsymbol{\Theta}$ needs to be first computed from the DTM. Therefore, during preprocessing, the height ground truth is converted into this consistent pixel-domain representation. For the entire SAR image

$$\mathbf{p}_{0,ij} = \mathbf{H}_{ij} / \boldsymbol{\rho}_{h,ij} \tag{8}$$

where the subscript ij represents the (i, j)th element of the corresponding matrix. $\mathbf{H}$ denotes the height ground truth matrix in the SAR image domain. $\mathbf{p}_0$ denotes the pixel count matrix, providing a scale-consistent representation of the ground truth.

Since the proposed method follows a depth-estimation formulation, the normalized height values are inverted to match the depth representation. In this way, ground-level pixels (0 m) are mapped to larger values, while higher structures correspond to smaller values. This is consistent with common practices in depth estimation, where larger values indicate greater distances from the sensor. In addition, the ground truth are normalized to the range $[-1, 1]$, following the convention of the original stable diffusion variational autoencoder (VAE) [25]. Accordingly, the normalized and scale-consistent ground truth used for supervised training is defined as $\mathbf{p}_{\text{norm}} = 1 - 2\mathbf{p}_0/M$, where M denotes the maximum estimable pixel count. In this study, M is set to 633, corresponding to the tallest building in the training SAR image, whose physical height is approximately 468 m. Moreover, 633 is sufficient to encompass the pixel counts of most buildings.

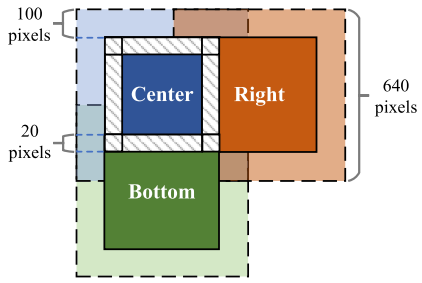


Fig. 6. Illustration of uniformly dividing a full SAR image into patches for testing, followed by cropping and stitching. The patch labeled "center" is shown as an example, along with its adjacent right and bottom patches. Patches are 640×640 pixels, with 100 pixels borders as invalid regions (lighter-colored areas) and 20 pixels overlaps (hatched areas) weighted to generate the final prediction.

C. Processing of Test Images

Test images are first subjected to the same radiometric calibration and normalization procedures as the training images. To further enhance the homogeneity of amplitude distributions between the training and test data, histogram matching is performed on the test images, using the histogram distribution of the training image as the reference. Since the test images have a resolution similar to that of the training images, a 2-times multilook processing is also applied to the test images. If the resolution of the test images differs significantly from that of the training images, it is recommended to first approximately unify their resolution before proceeding with prediction.

Subsequently, to enable inference over the entire SAR image, the full image is uniformly divided into patches that partially cover each other, which are independently fed into the network for prediction. Fig. 6 illustrates this process, and the details are described below. To obtain reliable estimation results, it is necessary to preserve the structural integrity of buildings within each test patch as much as possible. Following the strategy in [29], the edges of each patch, with a certain pixel width, are treated as an invalid region, and only predictions from the central area are used. In this study, since most buildings in the test set are below 150 m and the incidence angle ranges from 33° to 48°, (7) and (8) indicate that a margin of 100 pixels along the range direction is sufficient to cover the vast majority of buildings. Therefore, the width of the invalid border around each patch is set to 100 pixels. In addition, a 20-pixel overlap is applied to the central valid regions of adjacent patches. In the overlapping regions, pixels are distance-weighted with respect to patch centers and then combined to form the final result, thereby ensuring smooth transitions at patch edges. The patch size is

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set to 640 × 640 pixels, which is identical to the input size used during model training. It should be noted that, within the inference pipeline, all images are resampled to the optimal prediction size of 768 × 768 for performing inference, as this configuration provides the best prediction stability and accuracy for the stable diffusion model [25]. Therefore, the estimated absolute height values must be multiplied by a correction factor of 640/768 to obtain the correct heights.

During postprocessing of test results, the full-image local incidence angle matrix $\boldsymbol{\Theta}$ is first computed based on an external coarse DTM. In this study, for the GF-3 test images, LiDAR data are available, so the DTM is obtained from LiDAR-derived ground measurements. Using the parameter file of the test SAR image together with the local incidence angle matrix, the height pixel spacing matrix is then calculated. Finally, the estimated output values can be inversely converted to physical height values. The specific formulas are as follows:

$$\hat{\mathbf{H}}_{ij} = \hat{\mathbf{p}}_{ij} \cdot \frac{\rho_r}{\cos\left(\boldsymbol{\Theta}_{ij}\right)} \tag{9}$$

where the subscript ij represents the (i, j)th element of the corresponding matrix. $\hat{\mathbf{H}}$ denotes the predicted physical height matrix in the SAR image domain, and $\hat{\mathbf{p}}$ denotes the predicted pixel count matrix. The definitions of the remaining variables are the same as previously stated.

D. Data Augmentation

Under the extremely small dataset condition, essential data-augmentation strategies are required to enhance the model's generative capability. However, such strategies must be adopted with great caution. SAR images exhibit distinct scattering characteristics, and our experiments show that directly transferring conventional optical augmentations leads to unstable performance. Ultimately, two augmentation methods validated through ablation experiments were adopted. Both are applied during the random slicing of the full SAR image, namely, vertical flipping and dynamically scaling. The detailed procedures are described below.

1) Vertical Flipping: To increase the diversity of building shapes, vertical flipping (i.e., flipping along the azimuth direction) is performed with a probability of 0.5.

2) Dynamically Scaling: To account for building compression and stretching due to different range and azimuth resolutions and incidence angles, an image patch scaling step is introduced. Specifically, a larger sampling window is first extracted and then resampled to the size of the training patch, denoted as $[h, w]$, where w and h represent the width and height, respectively. It should be noted that, to avoid image distortion caused by oversampling of SAR images, the original inputs used in the experiments are required to be a single look image. This ensures that after sampling, the resizing operation only reduces the image size rather than enlarging it. Four sampling window size settings are considered, as described below.

Setting (I): As mentioned in Section II-A, the 2 times multilook image is used as the standard input for training and testing, so the first sampling window has a size of $[2h, 2w]$. After resampling to $[h, w]$, this window corresponds to the ground truth of the 2 times multilook SAR image.

Setting (II): To introduce more randomness and enhance the model's generalization, the second sampling window size is dynamic, being scaled up or down to varying degrees based on the standard window size. Specifically, the expansion coefficients c_1 and c_2 are independently sampled from a uniform distribution within the interval $[1.0, 3.0)$, resulting in a window size of $[c_1\ h, c_2 w]$. Extreme stretching can, to some extent, simulate changes in building morphology caused by different incident angles. However, experiments indicate that this dynamic sampling window may affect the stability of estimation, so its usage ratio is appropriately restricted.

Setting (III): Based on Setting II, the third sampling window setting further includes a sampling window of size $[h, w]$, with both window sizes weighted equally at 0.5. The reason is that experimental analysis shows that the original single look image can enhance the model's learning effect. This is presumably due to the fact that, although the original single look image has a limited receptive field, it contains the richest detail and texture information.

Setting (IV): To further improve the stability of the model's estimates, an attempt is made by combining the three sampling window sizes mentioned above. The windows $[2h, 2w]$, $[c_1\ h, c_2 w]$, and $[h, w]$ are assigned probabilities of 0.3, 0.3, and 0.4, respectively. In actual experiments, a random number within $[0, 1)$ is generated to select which sampling window size to use.

In addition, when multilook processing is applied to the range direction, the ground truth representing the number of pixels needs to be scaled accordingly, with the specific formula as follows:

$$\mathbf{p} = 1 - \left(1 - \mathbf{p}_{\text{norm}}\right) \times 2/c \tag{10}$$

where $\mathbf{p}_{\text{norm}}$ denotes the pixel count normalized to $[-1, 1]$, and $\mathbf{p}$ denotes the final dimensionless ground truth input to the model after scaling. c is the expansion coefficient of the dynamic range-direction window, $c \in [1.0, 3.0)$. For double multilook image, $c = 2$; for single-look image, $c = 1$; for dynamic window $[c_1\ h, c_2 w]$, $c = c_2$. It should be noted that the range of $\mathbf{p}$ has changed to $[-3, 1]$. The ground truth values within $[-3, -1)$ can be set to -1. However, to prevent gradient variation patterns from being affected by the saturation of the truth value, the values within $[-3, -1)$ are ultimately masked out during loss computation. Since tall buildings are relatively rare, this has little impact on the overall results.

III. Stable Diffusion-Based Model for SAR Height Estimation

This work trains a stable diffusion-based model to achieve task transfer from optical imagery to SAR imagery. Supported by a scale consistency algorithm, the diffusion model can be fine-tuned using only a very small SAR training dataset to estimate building heights from a single SAR image. The fine-tuning is performed in a supervised, end-to-end (E2E) manner, enabled by the findings in [27], which simplify the multistep denoising inference into a single-step process. Specifically, this study employs the E2E fine-tuned marigold model [25], [27].

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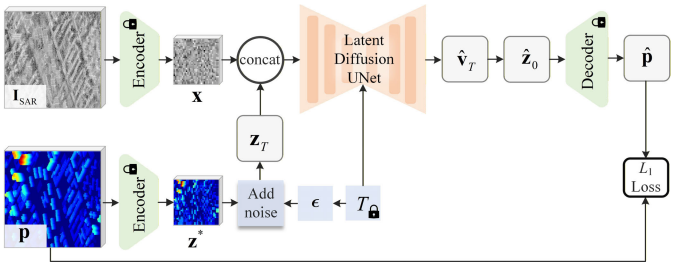


Fig. 7. Training workflow of the model for SAR building height estimation. The input is replaced with a small-sample dataset consisting of SAR amplitude images and their corresponding height maps. The pretrained parameters of the UNet in the E2E marigold model are fine-tuned. The lock icon indicates that this module is frozen, whose parameters remain unchanged.

The principles and implementation details of the model are described as follows.

A. Generation Formulation of the Model

In the diffusion model, the forward process gradually adds noise to the data. In the marigold, the noise is not added to the original data, but rather to the latent variable of the height map conditioned on the image, denoted as $\mathbf{z}^*$. As illustrated in Fig. 7, starting from the latent-space height map $\mathbf{z}^*$, the process applies noise addition over T time steps to obtain $\mathbf{z}_T$. Regarding the noise configuration, noise with variance β_t is added at each time step t, where β_t is a small predefined constant. After derivation, the relationship between $\mathbf{z}_T$ and the initial variable $\mathbf{z}^*$ in the forward process can be expressed as

$$\mathbf{z}_T = \sqrt{\bar{\alpha}_T}\mathbf{z}^* + \sqrt{1-\bar{\alpha}_T}\boldsymbol{\epsilon} \tag{11}$$

where $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_T = \prod_{t=1}^{T} \alpha_t$, $\boldsymbol{\epsilon}$ denotes the added noise. In the marigold model, the noise is sampled as $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \mathbf{I})$ [25], whereas in the optimized fine-tuned model, the noise is simplified to the mean of the distribution, i.e., 0 [27].

The noisy height map $\mathbf{z}_T$ and the SAR amplitude image $\mathbf{x}$ in the latent space are concatenated as $[\mathbf{z}_T, \mathbf{x}]$ and input into the UNet module along with the timestep T for the reverse process training. Marigold adopts a v-parameterized modeling objective, which is a combination of predicting the noise $\boldsymbol{\epsilon}$ and predicting the denoised image $\mathbf{z}^*$ through a linear transformation. Its optimization target $\mathbf{v}_T^*$ is

$$\mathbf{v}_T^* = \sqrt{\bar{\alpha}_T}\boldsymbol{\epsilon} - \sqrt{1-\bar{\alpha}_T}\mathbf{z}^*. \tag{12}$$

The number of steps T is set to a sufficiently large value, so $\bar{\alpha}_T \approx 0$, the model is trained to obtain a clean prediction $\mathbf{z}^*$ from pure noise via a single-step prediction. Essentially, the v-parameterization enables the model to predict the "velocity" of the denoising direction, which is numerically more stable than predicting $\boldsymbol{\epsilon}$ or $\mathbf{z}^*$ alone. Moreover, v-parameterization can be easily converted into a predicted clean image. By combining the relationship between the optimization target $\mathbf{v}_T^*$ and $\mathbf{z}^*$ in (11) and (12), the transformation relationship from the predicted value $\hat{\mathbf{v}}_T$ output by UNet to the predicted latent height map $\hat{\mathbf{z}}_0$ can be derived

$$\hat{\mathbf{z}}_0 = \sqrt{\bar{\alpha}_T}\mathbf{z}_T - \sqrt{1-\bar{\alpha}_T}\hat{\mathbf{v}}_\theta\left([\mathbf{z}_T, \mathbf{x}], T\right) \tag{13}$$

where

$$\hat{\mathbf{v}}_T = \hat{\mathbf{v}}_\theta\left([\mathbf{z}_T, \mathbf{x}], T\right) \tag{14}$$

and this formula specifically defines $\hat{\mathbf{v}}_T$ as the predicted output of the learned model for the concatenated input $[\mathbf{z}_T, \mathbf{x}]$ and time step T. Subsequently, $\hat{\mathbf{z}}_0$ is decoded via the frozen VAE decoder to obtain the predicted map $\hat{\mathbf{p}}$, which is then compared with the ground truth map $\mathbf{p}$. The loss is computed according to the absolute height loss function and backpropagated to update the trainable parameters, enabling E2E training.

B. Model Architecture

The stable diffusion model operates in the latent space of a VAE. Therefore, both the input image and the conditional image are first passed through the encoder $\mathcal{E}$, and the predicted output in the latent space is finally decoded via the decoder $\mathcal{D}$ to obtain the predicted depth map. The VAE is trained independently of the LDM, with the objective that the encoder $\mathcal{E}$ compresses the input into a low-dimensional latent variable and the decoder $\mathcal{D}$ reconstructs the input, i.e., $\mathcal{D}(\mathcal{E}(\mathbf{x})) \approx \mathbf{x}$. In this study, we use the original stable diffusion VAE, which is kept frozen during training.

To comply with the input format of the marigold model for RGB images, the single-channel SAR amplitude map is duplicated to form a three-channel image, denoted as $\mathbf{I}_{\text{SAR}}$. After passing through the encoder, the latent representation of the SAR image is obtained as $\mathbf{x} = \mathcal{E}(\mathbf{I}_{\text{SAR}})$. The SAR height map $\mathbf{p}$ has the same dimensions as the optical depth maps used in the original model and can therefore be directly converted to latent variables: $\mathbf{z}^* = \mathcal{E}(\mathbf{p})$.

During model training, only the denoising UNet is fine-tuned, which has been shown to be sufficient to leverage the potential of the pretrained diffusion model [25]. Regarding the feasibility of single-step inference, a brief summary is provided here (see [27] for details). In conditional diffusion, both the forward and backward processes are conditioned on the ground truth image. The network is trained to denoise given the conditional image. The optimal prediction at the final time step T corresponds to the mean of the data distribution [41]. For image-conditioned generation, since each image corresponds to a single depth map, the optimal prediction at T is expected to closely approximate the ground-truth depth map.

C. Loss Function Design

Unlike the affine-invariant relative depth loss commonly used for optical images [25], [42], for the task of building height estimation from SAR images, the model's target is converted to the number of pixels covered by buildings. This formulation is more consistent with absolute depth estimation. The L_1 loss is adopted as the loss function. Experiments were also conducted using L_2 loss and with additional gradient-based losses; however, the results were either inferior to or comparable with the simple L_1 loss. Therefore, L_1 loss was ultimately chosen.

In addition, since the valid height range is set to $[-1, 1]$ to match the input range of the VAE, values that fall outside

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TABLE I
PARAMETERS OF THE GF-3 SAR IMAGE DATASETS USED IN THIS STUDY

City	Abbr.	Satellite	Orbit Number	Mode	Pol.	Orbit	Incidence Angle (°)	SAR Acquisition Date	Range Pixel Spacing × Azimuth Pixel Spacing (m)	For Training	LiDAR Acquisition Year
New York City, USA	NYC_1	GF-3	019605	SL	HH	A	40.4	2020.04	0.56*0.34	✓	2017
	NYC_2	GF-3	017939	SL	HH	D	37.9	2020.01	0.56*0.33	✗	
	NYC_3	GF-3	018011	SL	HH	D	45.6	2020.01	0.56*0.34	✗	
San Diego, USA	San_1	GF-3	034849	SL	HH	D	42.8	2023.03	0.56*0.34	✗	2014
	San_2	GF-3	042720	SL	HH	D	32.7	2024.09	0.56*0.34	✗	
	San_3	GF-3	035021	SL	HH	D	38.4	2023.04	0.56*0.33	✗	
	San_4	GF-3	027325	SL	HH	D	42.3	2021.10	0.56*0.33	✗	
Norfolk, USA	Nor_1	GF-3	035322	SL	HH	D	48.3	2023.04	0.56*0.36	✗	2013
	Nor_2	GF-3	035005	SL	HH	D	36.4	2023.04	0.56*0.33	✗	
Washington, D.C., USA	Wash_1	GF-3	993931	SL	HH	A	38.4	2025.05	0.56*0.33	✗	2022

Note: Abbr. denotes abbreviation. Pol. denotes polarization.

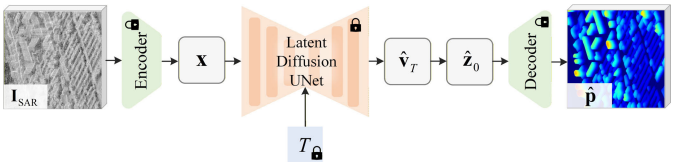


Fig. 8. Model inference workflow. The lock icon indicates that this module is frozen, whose parameters remain unchanged.

this range due to scaling transformations (see Section II-D) are masked in the loss function. The mask matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$ is defined as

$$\mathbf{M}_{ij} = \begin{cases} 0, & \text{if } \mathbf{p}_{ij} < -1 \\ 1, & \text{otherwise.} \end{cases} \tag{15}$$

For the ground truth and predicted maps, $\mathbf{p}, \hat{\mathbf{p}} \in \mathbb{R}^{m\times n}$, the masked loss function is then defined as

$$\mathcal{L}(\mathbf{p}, \hat{\mathbf{p}}, \mathbf{M}) = \frac{1}{\sum_{i=1}^{m}\sum_{j=1}^{n}\mathbf{M}_{ij}} \sum_{i=1}^{m}\sum_{j=1}^{n}\mathbf{M}_{ij}\left|\mathbf{p}_{ij} - \hat{\mathbf{p}}_{ij}\right|. \tag{16}$$

D. Inference Phase

The inference procedure is illustrated in Fig. 8 and is deterministic. The SAR image $\mathbf{I}_{\text{SAR}}$ is first encoded into the latent space via the encoder. The resulting latent variable $\mathbf{x}$ and time steps T are then processed by the latent diffusion UNet, using the parameters learned during training, to produce the predicted latent output $\hat{\mathbf{v}}_T$. This output is subsequently transformed into the latent height prediction $\hat{\mathbf{z}}_0$, which is decoded by the VAE decoder $\mathcal{D}$ to generate the final predicted building height map $\hat{\mathbf{p}}$.

E. Implementation Details

The officially released checkpoint of the E2E fine-tuned marigold model with the best overall performance in [27] was adopted. The model was optimized using the AdamW optimizer for 2000 iterations. The number of iterations was significantly smaller than that in the original design because the dataset used in this study is much smaller than the optical dataset in the original work. Accordingly, the initial learning rate was adjusted from 3×10^{-5} to 5×10^{-5}, followed by an exponential decay after a warm-up period of 100 steps, consistent with the original setup. Since only a single dataset is used, data fusion among multiple datasets is unnecessary, and the batch size can therefore be set to 1. To enable training with relatively large input image patches of 640×640 pixels on a single NVIDIA V100S GPU, the batch size was fixed at 1, which occupied approximately 28.9 GB of GPU memory. Gradient accumulation was applied to maintain an effective batch size of 32. The fine-tuning process took approximately 3.5 h on eight NVIDIA V100S GPUs, for which the corresponding accumulation steps were set to 4.

IV. EXPERIMENTS

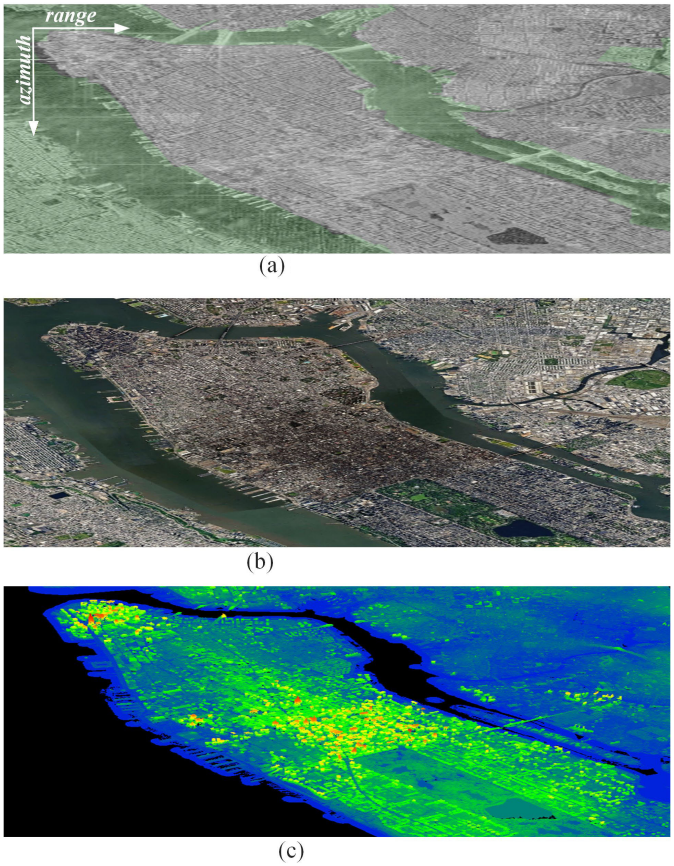
A. Dataset Description

1) Training Dataset: A GF-3 SAR image of the NYC area from the SL model was selected as the training dataset, with detailed parameters listed in the first row of Table I. A data thumbnail of the region is shown in Fig. 9. Among the available data, the NYC area features dense building distributions with various heights; therefore, it was chosen as the training dataset.

A dataset comprising 640×640 pixels images was generated from regions with available ground truth within the single SAR image covering the NYC area. To ensure full coverage, the total number of patches was calculated to be three times the number required to cover all pixels. The total number of valid pixels is 47 162 565, resulting in 345 image patches. During training, 2000 iterations were performed, with 32 image patches (corresponding to the effective batch size after gradient accumulation) input per iteration, resulting in a total of 182 training epochs.

2) Testing Dataset: All data entries in Table I, except for the three images from NYC, were used as the test set.

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(a)

(b)

(c)

Fig. 9. Comparative thumbnails of the SAR image, optical image (Google Earth, Image 2026 Airbus), and LiDAR point cloud data in the training area. (a) Green area: invalid mask region, comprising areas lacking ground truth labels, water bodies, and other nonbuilding structures. (a) SAR image. (b) Optical image. (c) LiDAR.

These regions are entirely unseen by the model and cover SAR images with varying resolutions and incidence angles. The test set consists of seven GF-3 images acquired in SL mode from three different cities. The test dataset includes San Diego, USA, characterized by many high-rise buildings between 100–150 m; Washington, DC, USA, characterized by predominantly low-rise buildings below 55 m; and Norfolk, USA, characterized by the tallest buildings of approximately 90 m, with most structures being low-rise and heavily vegetated.

B. Quantitative Evaluation Metrics

All quantitative evaluations in this study are performed on the building heights in the SAR image domain. Therefore, before evaluation, the predicted dimensionless value $\hat{\mathbf{p}}$ is first converted into the predicted physical height value $\hat{\mathbf{H}}$, as described in Section II-C. This conversion ensures that the evaluation metrics have clear physical significance.

Two metrics are employed for evaluation. The first is the mean absolute error (MAE), which represents the average absolute difference between the predicted and ground-truth values, and is defined as follows:

$$\mathrm{MAE} = \frac{\sum_{i=1}^{N} \left|\hat{\mathbf{H}}_i - \mathbf{H}_i\right|}{N}. \tag{17}$$

Here, $\mathbf{H}, \hat{\mathbf{H}} \in \mathbb{R}^{m\times n}$ represent the ground-truth and predicted height maps, respectively, and $N = m \times n$ is the total number of pixels, with the subscript i referring to the ith pixel.

The second evaluation metric is the root-mean-square error (RMSE), which is more sensitive to larger errors. It is defined as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{\mathbf{H}}_i - \mathbf{H}_i\right)^2}. \tag{18}$$

The variables have the same meanings as those in the MAE definition.

C. Comparison of Different Models

To demonstrate the effectiveness and unique advantages of the proposed method used in this study for SAR image building height estimation, three baselines were selected for comparison: a standard UNet, the state-of-the-art SAR2Height network [29] in this field, and METER [43], a vision transformer model originally designed for monocular optical image depth estimation. Both UNet and METER were trained using a single SAR image, consistent with the diffusion model. In contrast, SAR2Height, which incorporates an incidence angle module, was first trained on three NYC images [denoted as S2H (three imgs)]. To better reproduce its performance observed in large-scale training, a second version was trained on six images [denoted as S2H (six imgs)]. All trained baselines and the proposed model were tested on seven previously unseen SAR images from three cities. For the test SAR images from the San Diego region, S2H (six imgs) was trained using all remaining images from the other cities as the training dataset. For the images from the Norfolk and Washington, DC, regions, S2H (six imgs) was trained using three images from NYC and the first three images from San Diego.

Since the UNet and METER models lack scale consistency, manual correction is performed. Two representative points are selected from the prediction map for a least-squares fit to obtain an approximate scaling factor k. The predictions are then corrected as $h_{\mathrm{corr}} = \hat{h} \cdot k$ before conducting qualitative and quantitative analyses.

Quantitative comparisons in terms of MAE and RMSE over seven test images from three cities are reported in Fig. 10 and Table II. All quantitative results were computed over dense building areas where water bodies and vegetation regions were largely excluded. The MAE results indicate that the estimation error increases with the overall building height, following the order (lowest to highest): Washington < Norfolk < San Diego. This is mainly because layover effects are more complex in high-rise urban areas, and even small estimation biases can lead to noticeable metric degradation. A similar trend can also be observed in the RMSE results. Compared with the baseline methods, the proposed approach achieves the best estimation performance, indicating its effectiveness in transferring optical prior information to SAR imagery.

In addition, qualitative evaluations are performed on SAR test scenes from a low-rise region (Washington, DC, USA) and a high-rise region [San Diego, USA)]. The estimated results of *Wash_1* are shown in Fig. 11, and the corresponding height profiles extracted from local patches in dense building

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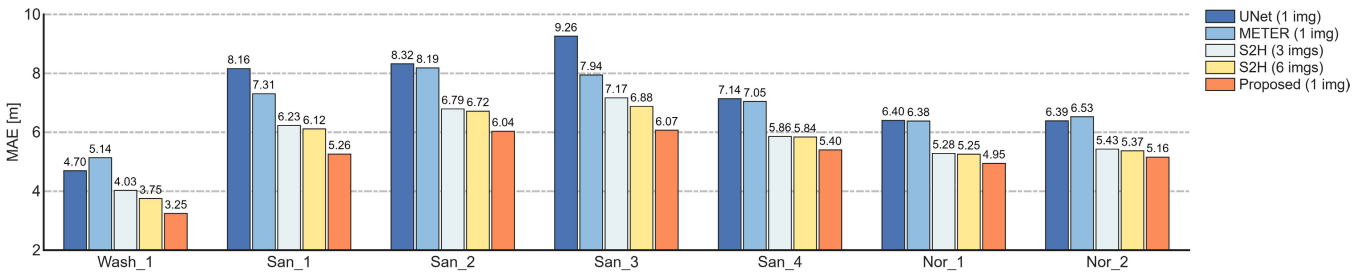


Fig. 10. Comparison of MAE for test results of different models on seven unseen SAR images within dense building regions.

TABLE II
RMSE RESULTS OF DIFFERENT MODELS ON PREVIOUSLY UNSEEN SAR TEST IMAGES

Test Image	RMSE (m)					Gain (%)
	UNet (1 img)	METER (1 img)	S2H (3 imgs)	S2H (6 imgs)	Proposed (1 img)	
Wash_1	8.42	9.09 ↑	7.57 ↓	6.91 ↓	**5.96** ↓	13.75
San_1	16.50	13.81 ↓	13.06 ↓	12.37 ↓	**10.88** ↓	12.05
San_2	17.61	16.67 ↓	14.87 ↓	14.76 ↓	**12.89** ↓	12.67
San_3	18.01	15.20 ↓	14.73 ↓	13.79 ↓	**12.24** ↓	11.24
San_4	13.95	15.20 ↑	12.33 ↓	11.67 ↓	**11.08** ↓	5.06
Nor_1	11.11	11.08 ↓	9.50 ↓	9.48 ↓	**8.95** ↓	5.59
Nor_2	11.36	11.45 ↑	9.93 ↓	9.83 ↓	**9.42** ↓	4.17

Note: Lower RMSE indicates higher accuracy. Bold indicates the best result, and underline indicates the second-best result. Gain (%) denotes the relative improvement of the proposed method over the best baseline. ↑ and ↓ denote an increase and a decrease relative to the preceding method, respectively.

areas are shown in Fig. 12. The estimated results of *San_1* are shown in Fig. 13, with the corresponding height profiles presented in Fig. 14. Image patches of 512 × 512 pixels were used to examine representative local structures at a higher resolution. The performance of UNet and METER using a single image is relatively poor. In contrast, SAR2Height exhibits improved performance as the number of input images increases from 3 to 6. Using only a single image, the proposed method shows the best overall performance in terms of building structural integrity and height accuracy. It is also effective in handling complex layover effects in high-rise buildings exceeding 100 m in San Diego. The proposed method yields profiles that are closer to the ground truth and better preserve the structural consistency of height variations.

D. Cross-Sensor Evaluation on TerraSAR-X Data

To further evaluate the proposed method under a cross-sensor setting, two TerraSAR-X images are tested: staring spotlight (ST) over Dublin, Ireland, and high-resolution spotlight (HS) over Rotterdam, The Netherlands. It should be noted that GF-3 operates in the C-band, whereas TerraSAR-X operates in the X-band. The range pixel spacing of both images is approximately 0.45 m, while the azimuth spacing is 0.16 m for ST and 0.86 m for HS. First, resampling is performed to make the azimuth and range pixel spacing's approximately equal, thereby avoiding excessive stretching of building geometry. The resampled spacing is selected mainly to make full use of the available data. In the experiments presented here, the ST image is resampled to approximately 0.5 m, whereas the HS image is resampled to approximately 1 m. The subsequent procedure follows the Testing pipeline illustrated in Fig. 1. During data preprocessing, in order to minimally distort the SAR information, only the amplitude of the TerraSAR-X images is converted to a logarithmic (dB) scale, and the main interval of $[0, 150]$ dB is normalized to [0, 1]. The data are then fed into the network for prediction. In postprocessing, the public SRTM 90 m DEM is used as a coarse DTM.

The results are shown in Fig. 15, where in (d) and (h) the white letter markers in the prediction map are quantitatively compared with the corresponding ground truth from the 3-D optical map in Google Earth Pro. Accurate height estimation is achieved for most building structures, with errors generally within or close to the input image pixel spacing, indicating effective pixel-level estimation. The ST mode achieves higher estimation accuracy than the HS mode, suggesting that better SAR image quality is beneficial to height estimation. However, the building heights marked by the three black numerical labels are not successfully estimated. These buildings exhibit mainly edge-like outlines with no prominent side facades in the SAR images, which likely increases the reconstruction difficulty.

E. Ablation Study

1) Number of Iterations: The number of training iterations affects the performance of the diffusion model. A total of 5000 iterations were used, with the learning rate warmed up for 100 steps to 5×10^{-5}, and then exponentially decayed to 0.01 of its initial value. Owing to the large fluctuation of the per-step MAE loss, the mean loss per epoch is reported for training loss visualization. As shown in Fig. 16, the training loss gradually converges.

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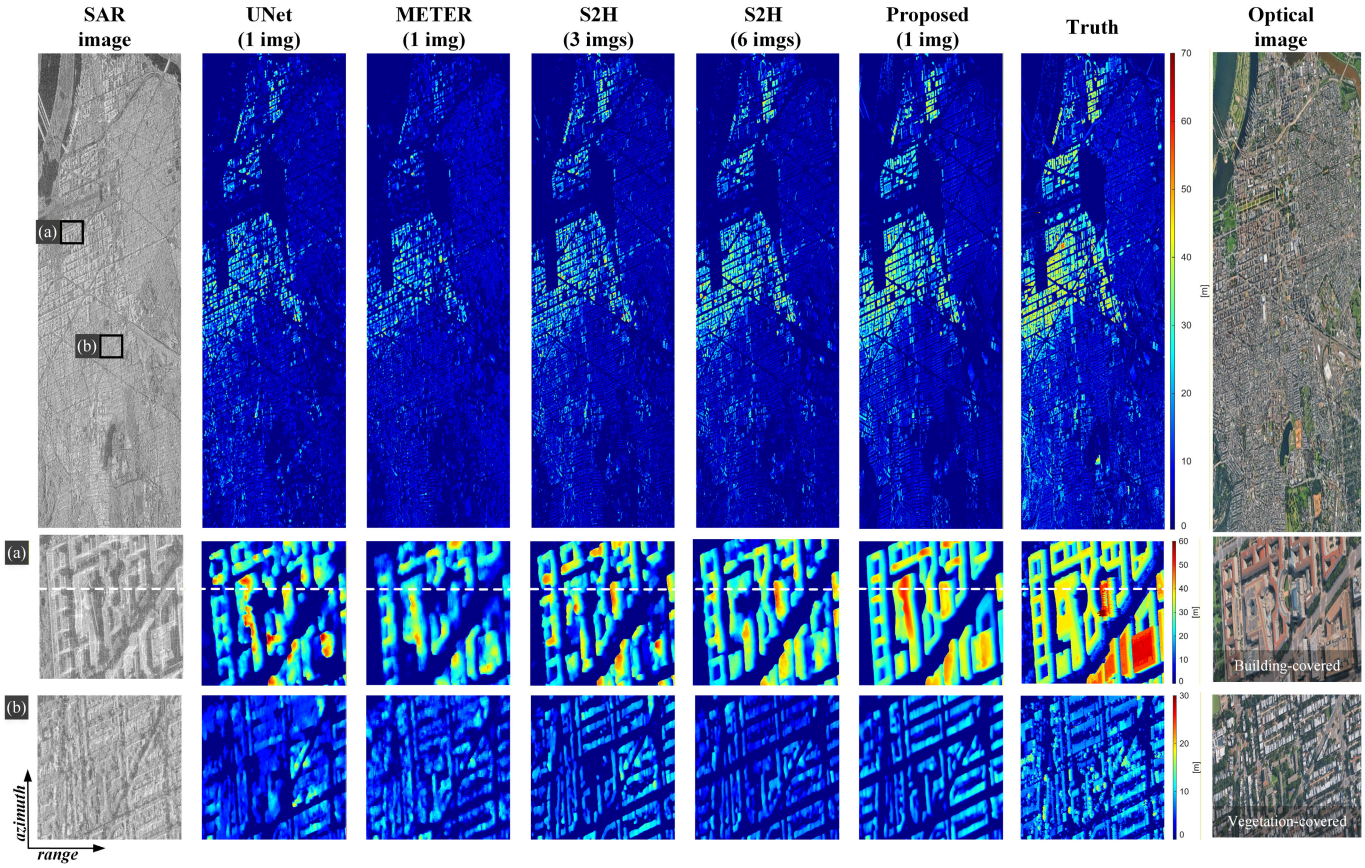


Fig. 11. Comparison of four models on the previously unseen *Wash_1* SAR test image. "one img," "three imgs," and "six imgs" denote the number of SAR images used for training. The baselines include the standard UNet, METER [43], and S2H (SAR2Height) [29] (a) and (b) indicate two representative local regions enlarged for detailed comparison. The UNet, METER, and the proposed method are all trained using a single SAR image, while SAR2Height is trained with three SAR images acquired at different incidence angles to enable its incidence angle module. These settings are designed to simulate tiny dataset training conditions. To better reproduce the performance of SAR2Height, results obtained with enhanced training using six SAR images from nontest regions are also shown. (a) Height profile along the white line is compared in Fig. 12. Missing-value regions in the ground truth are masked out. Optical images are from Google Earth, Image 2026 Airbus.

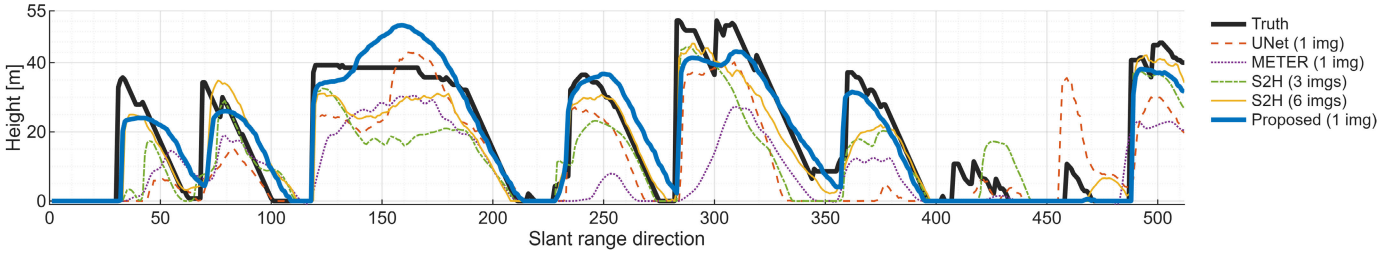


Fig. 12. Height profile contrast in Fig. 11 white dashed line.

To compute the testing loss, checkpoints were saved every 1000 iterations during training. For each checkpoint, inference was performed on seven previously unseen images, and the MAE loss was computed (see Table III). The results indicate that, across different test images, the MAE varies only slightly with training iterations, and the minimum MAE is achieved at 2000 iterations. To evaluate the ablation performance across different building height ranges, two representative test images, *Wash_1* and *San_1*, were selected, corresponding to low-rise and high-rise building areas, respectively, with their MAE results visualized in Fig. 16. Specifically, *Wash_1* mainly contains buildings with heights below 55 m, whereas *San_1* includes several buildings in the 100–150-m height range. Note that in the ablation experiments, the MAE loss was computed only within masked dense building regions, in order to ensure a fair and focused evaluation of building height estimation accuracy.

Based on the above observations, 2000 iterations were selected as the final training setting. The learning rate schedule remained unchanged.

2) Data Augmentation Strategies: Since data augmentation was applied to the small dataset, it is necessary to perform an ablation study on the augmentation strategies. The different

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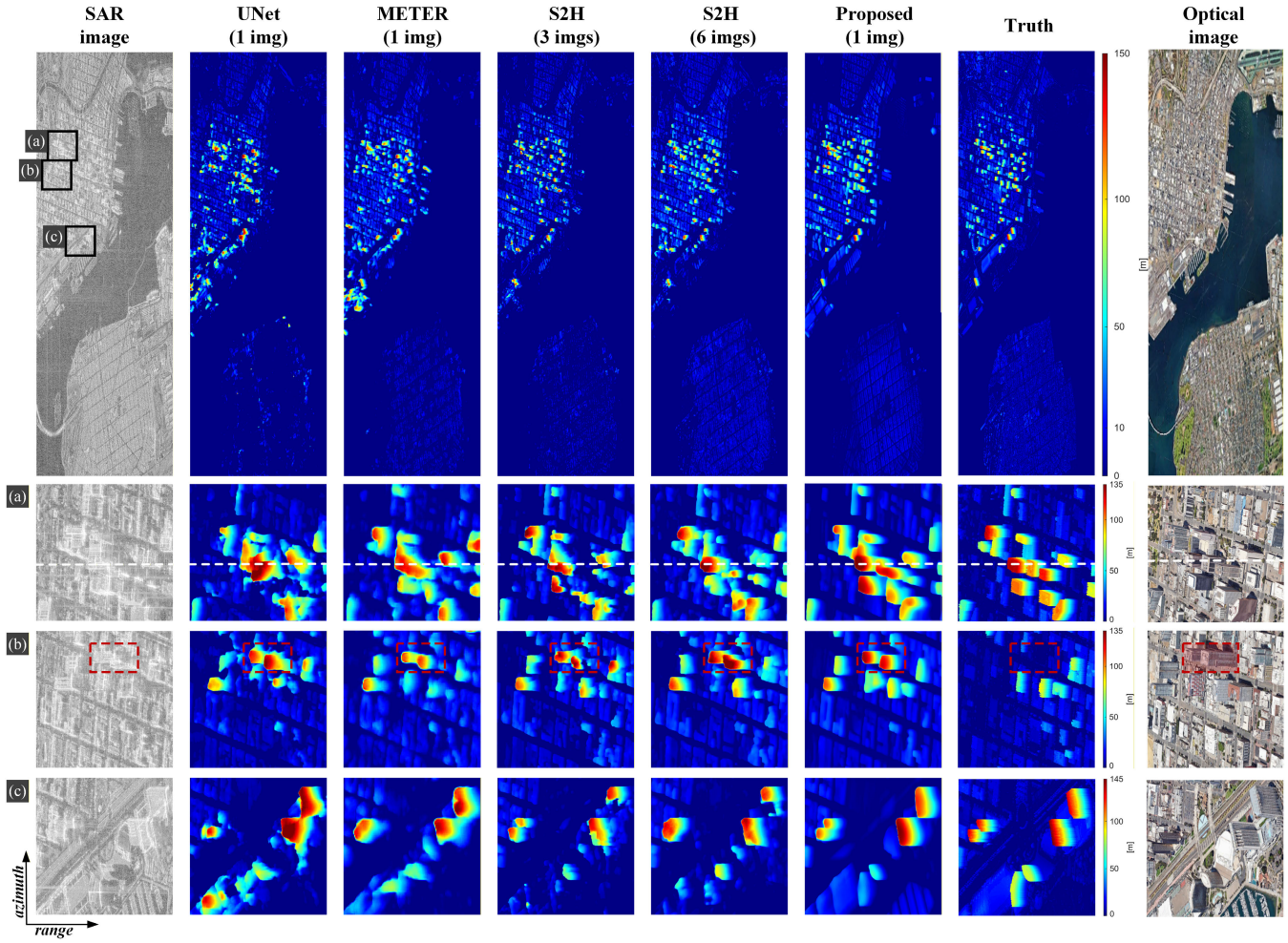


Fig. 13. Comparison of four models on the *San_1* SAR image (unseen during training). S2H (three imgs) indicates that SAR2Height [29] was trained using three SAR images from the NYC dataset, which still represents a tiny-dataset setting for this method (a)–(c) indicate three representative local regions enlarged for detailed comparison. The ocean areas and the missing value regions in the ground truth are masked out. (a) Height profile along the white line is compared in Fig. 14. (b) Red-boxed area shows no buildings in the LiDAR ground truth, whereas the optical 3-D map reveals two buildings with the highest points of about 133 m and lower points of about 122 m, which is consistent with the estimates of the proposed method. Optical images are from Google Earth, Image 2026 Airbus.

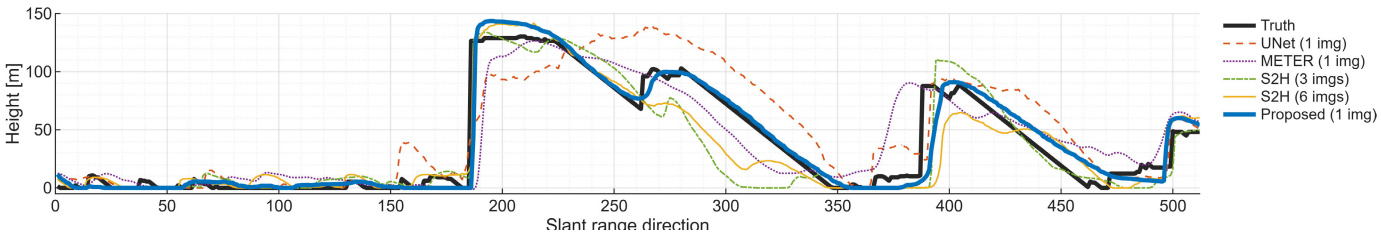
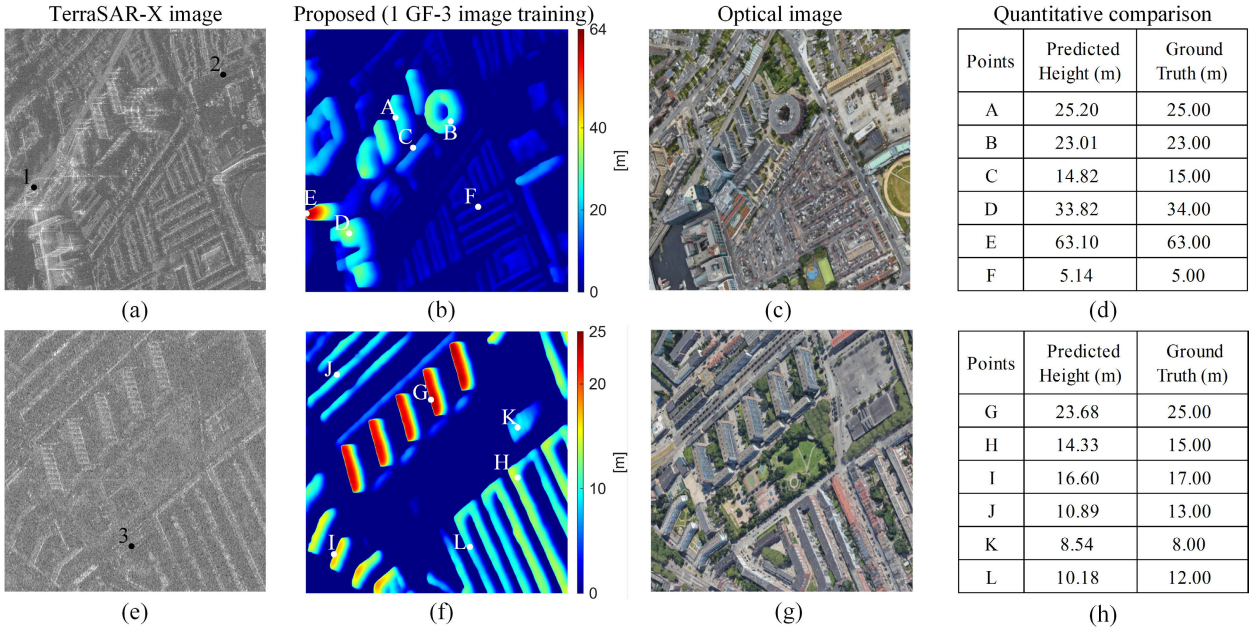


Fig. 14. Height profile contrast in Fig. 13 white dashed line.

strategies and their combinations are listed in Table IV. By sequentially adding each augmentation technique and conducting qualitative and quantitative comparisons of the estimated height results in the same test region, the necessity and effectiveness of the strategies are verified. Fig. 17 shows a qualitative comparison using two slice examples of low-rise and high-rise buildings. For augmentation versions V1–V5, Fig. 18 presents the training MAE loss curves, and Table V reports the test MAE in building-dense areas. According to Table V, the V4 model version exhibits the best overall performance, while the V5 version ranks second, with both showing comparable estimation accuracy. This result indicates that a tradeoff is required when selecting dynamic scaling sizes, and Setting (III) provides a relatively optimal configuration. Together with the training loss curves in Fig. 18, it can be observed that with progressively stronger data augmentation, the training MAE increases, whereas the test MAE consistently decreases. This behavior is characteristic of effective regularization, indicating that the proposed augmentation strategy mitigates overfitting and enhances generalization performance

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Points	Predicted Height (m)	Ground Truth (m)
A	25.20	25.00
B	23.01	23.00
C	14.82	15.00
D	33.82	34.00
E	63.10	63.00
F	5.14	5.00

Points	Predicted Height (m)	Ground Truth (m)
G	23.68	25.00
H	14.33	15.00
I	16.60	17.00
J	10.89	13.00
K	8.54	8.00
L	10.18	12.00

Fig. 15. Test results of the proposed framework on TerraSAR-X data, trained with a single GF-3 SAR image, for cross-sensor evaluation. The first and second rows correspond to the TerraSAR-X ST and HS imaging modes, respectively. (a)–(d) show the SAR image, the predicted height map, the optical image (Google Earth, Image 2026 Airbus), and the quantitative comparison of the marked points for the ST mode, respectively. (e)–(h) show the corresponding results for the HS mode. Ground truth values are obtained from the 3-D map in Google Earth Pro. White lettered points A–L indicate quantitative comparison locations, and black numerical labels 1–3 denote positions where heights were not successfully estimated.

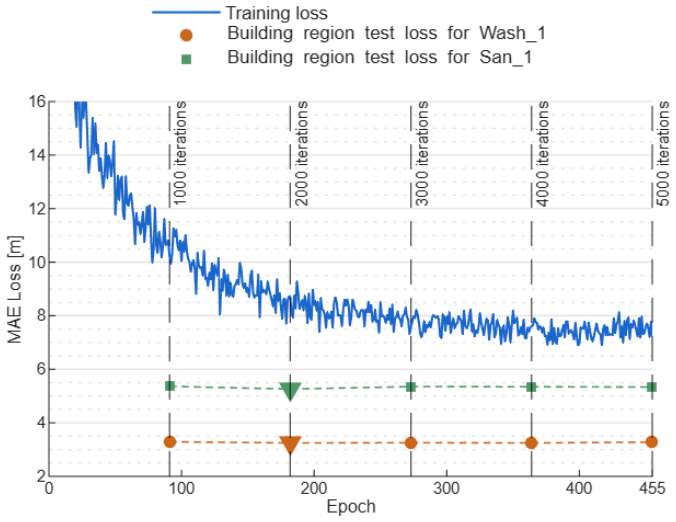


Fig. 16. Step ablation results over 5000 iterations (455 epochs), showing average training MAE per epoch. Checkpoints are saved every 1000 iterations. Test losses in dense building areas of two previously unseen images (*San_1*: taller buildings; *Wash_1*: shorter buildings) are shown. Minimum test losses at 2000 iterations are marked with inverted triangles.

TABLE III
MAE RESULTS OF THE PROPOSED MODEL ON PREVIOUSLY UNSEEN SAR TEST IMAGES AT DIFFERENT TRAINING ITERATIONS

Test Image	MAE (m)				
	1000	2000	3000	4000	5000
Wash_1	3.292	**3.248**	3.254	3.248	3.276
San_1	5.362	**5.259**	5.352	5.346	5.335
San_2	6.078	6.035	6.028	**5.976**	5.996
San_3	6.189	**6.070**	6.074	6.071	6.073
San_4	8.364	**8.174**	8.309	8.315	8.281
Nor_1	**4.926**	4.948	4.989	4.977	4.948
Nor_2	**5.091**	5.156	5.153	5.149	5.145
Average	5.615	**5.556**	5.594	5.583	5.579

TABLE IV
MODEL VERSIONS TRAINED WITH COMBINED DATA AUGMENTATION STRATEGIES

Model Version	Original Image	Vertical Flip	Dynamic Scale (Multilook Setting)			
			Setting (I)	Setting (II)	Setting (III)	Setting (IV)
V1	✓	✗	✓	✗	✗	✗
V2	✓	✓	✓	✗	✗	✗
V3	✓	✓	✗	✓	✗	✗
V4	✓	✓	✗	✗	✓	✗
V5	✓	✓	✗	✗	✗	✓

Note: Sample multilook settings:
(I) 2 times multilook;
(II) 1–3 times multilook;
(III) 1 time and 1–3 times multilook, each with a weight of 0.5;
(IV) 1 time, 1–3 times, and 2 times multilook, with weights of 0.4, 0.3, and 0.3, respectively.

in complex urban scenes. The results presented in the other parts of the article are all obtained using the V4 version of the model.

3) Scale Consistency Module: To quantitatively validate the effectiveness of the SCM, an ablation study is conducted by removing this module. In this setting, the ground-truth representation is replaced from the unified scale space with the original physical height, while all other training configurations remain unchanged. The models are evaluated on the same unseen test images, and the MAE is computed for comparison. The results are reported in Table VI. The effect of the SCM

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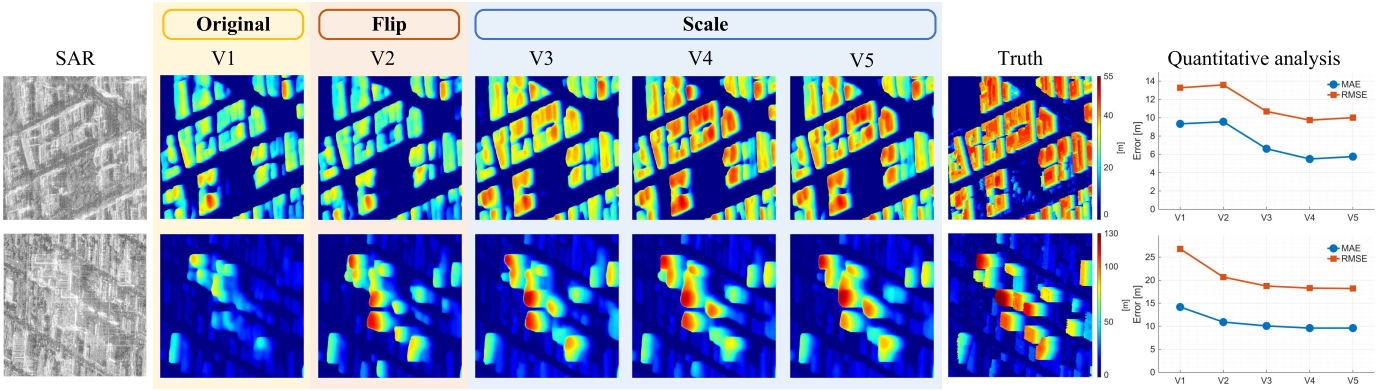


Fig. 17. Comparison of example slices from test images under different data augmentation strategies (V1–V5, see Table IV). The first row shows a slice from *Wash_1* (low-rise area). The second row shows a slice from *San_2* (high-rise area).

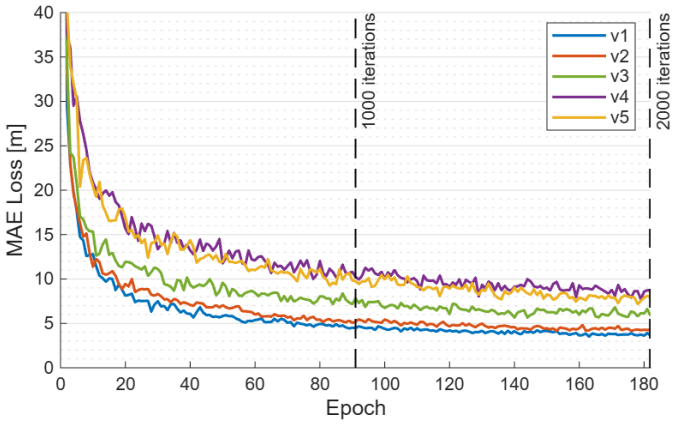


Fig. 18. Comparison of training MAE curves under different data augmentation versions (V1–V5) during 2000 iterations (182 epochs).

TABLE V
MAE RESULTS OF DIFFERENT MODEL VERSIONS (LISTED IN TABLE IV) ON UNSEEN TEST SAR IMAGES

Test Image	MAE (m)				
	V1	V2	V3	V4	V5
Wash_1	4.10	4.28	3.49	3.25	**3.22**
San_1	6.72	5.64	5.79	**5.26**	5.59
San_2	7.55	6.54	**6.01**	6.04	6.07
San_3	7.44	6.52	6.27	**6.07**	6.13
San_4	8.71	8.24	8.46	8.17	**8.15**
Nor_1	5.50	5.33	5.33	**4.95**	5.00
Nor_2	5.56	5.75	5.51	5.16	**5.17**
Average	6.51	6.04	5.84	**5.56**	5.62

Note: Bold indicates the best performance, while underlining denotes the second-best performance.

TABLE VI
ABLATION STUDY OF THE SCM

Test Image	MAE (m)		MAE Reduction (%)
	Proposed w/o SCM	Proposed	
Wash_1	4.37	**3.25**	25.6
San_1	5.54	**5.26**	5.1
San_2	6.73	**6.04**	10.3
San_3	6.76	**6.07**	10.2
San_4	8.31	**8.17**	1.7
Nor_1	5.39	**4.95**	8.2
Nor_2	5.93	**5.16**	13.0
Average	6.15	**5.56**	9.6

Note: MAE Reduction (%) is computed as $(\mathrm{MAE}_{\text{w/o SCM}} - \mathrm{MAE}_{\text{Proposed}})/\mathrm{MAE}_{\text{w/o SCM}} \times 100\%$.

TABLE VII
ABLATION STUDY OF GROUND-TRUTH INVERSION

Test Image	MAE (m)		MAE Reduction (%)
	Proposed w/o GT inversion	Proposed	
Wash_1	4.07	**3.25**	20.2
San_1	5.27	**5.26**	0.2
San_2	6.50	**6.04**	7.1
San_3	6.36	**6.07**	4.6
San_4	8.30	**8.17**	1.6
Nor_1	5.00	**4.95**	1.0
Nor_2	5.69	**5.16**	9.3
Average	5.88	**5.56**	5.4

Note: "w/o GT inversion" denotes training without ground-truth inversion. MAE Reduction [%] is computed as $(\mathrm{MAE}_{\text{w/o GT inversion}} - \mathrm{MAE}_{\text{Proposed}})/\mathrm{MAE}_{\text{w/o GT inversion}} \times 100\%$.

varies across test images, mainly due to differences in height pixel spacing between the training and test data, which are largely induced by differences in the incidence angle.

4) Ground-Truth Value Inversion: To validate the ground-truth value inversion, an ablation study is conducted under identical settings, with and without inverting the normalized ground truth. MAE is evaluated on multiple unseen test images. As shown in Table VII, inversion consistently improves the estimation accuracy, confirming that the inverted representation is better suited to the depth-estimation formulation.

5) Vegetation-Removed Ground Truth: It was observed in the experiments that the presence of vegetation in the ground truth has a significant impact on the model's estimation performance. This phenomenon can be understood by observing SAR images and their corresponding ground truth in vegetation-rich areas, as shown in Fig. 19. In SAR images, the distinction between trees and buildings is quite pronounced. Buildings typically exhibit geometric patterns of

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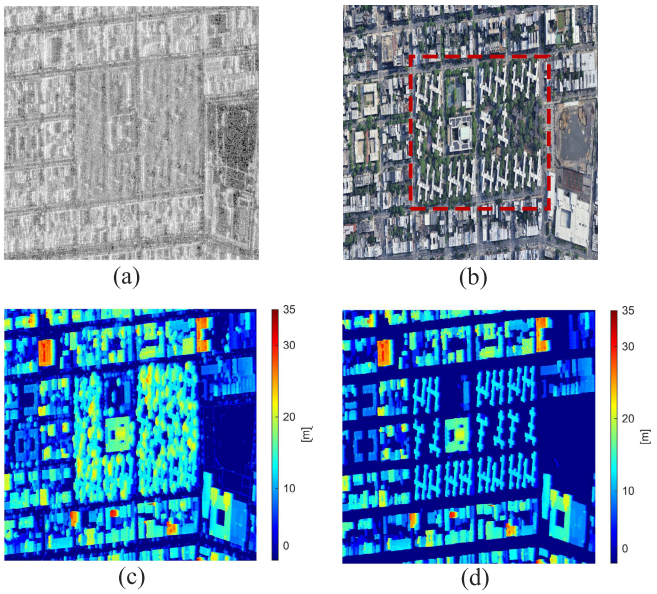


Fig. 19. Example of closely adjacent vegetation and buildings (a) SAR image, (b) optical image (Google Earth, Image 2026 Airbus), (c) ground truth with vegetation, and (d) ground truth with only buildings. The red box in (b) indicates areas of dense vegetation.

TABLE VIII

MAE COMPARISON OF MODELS TRAINED WITH/WITHOUT VEGETATION IN GROUND TRUTH ON UNSEEN TEST SAR IMAGES

Test Image	MAE (m) (Training with Ground Truth)		MAE Reduction (%)
	With Vegetation	Without Vegetation	
Wash_1	3.55	**3.25**	8.5
San_1	5.51	**5.26**	4.5
San_2	6.41	**6.04**	5.8
San_3	6.55	**6.07**	7.3
San_4	8.27	**8.17**	1.2
Nor_1	5.03	**4.95**	1.6
Nor_2	5.56	**5.16**	7.3
Average	5.84	**5.56**	5.2

Note: MAE Reduction (%) is computed as $(\mathrm{MAE}_{\mathrm{with}} - \mathrm{MAE}_{\mathrm{without}})/\mathrm{MAE}_{\mathrm{with}} \times 100\%$.

strong scattering points arising from their surface materials, which provide important cues for height estimation. In contrast, vegetation tends to produce blurred scattering structures, which makes the learning of vegetation-related ground truth more challenging for the model. In particular, when vegetation height is comparable to or exceeds that of buildings, the building ground truth may be obscured, as shown in Fig. 19(c).

The ablation study is set up as follows. The two NYC region datasets, with and without vegetation, were both trained under identical settings. The MAE is computed on seven previously unseen test images for comparison, as summarized in Table VIII. On average, removing vegetation from the ground truth reduces the MAE by 5.2%. A visual comparison and scatter plot analysis of the estimation results over the building-dense region in *San_2* are presented in Fig. 20. The use of vegetation-included ground truth leads to blurred building boundaries [Fig. 20(c)] and degrades the estimation of

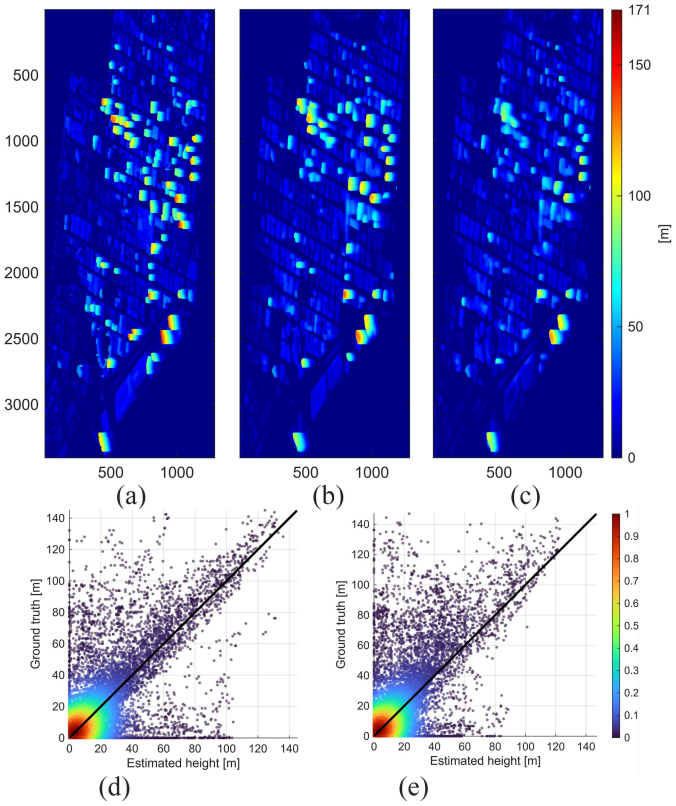


Fig. 20. Example from the *San_2* scene illustrating the effect of including vegetation in the ground truth. (a)–(c) ground truth, and the test results without and with vegetation, respectively. (d) and (e) 2-D scatter plots comparing (b) and (c) with the ground truth (a), using 30 000 randomly sampled points. In (d), the scatter points are largely symmetric around the ground truth and closely follow it. In (e), the predictions tend to underestimate the ground truth. This comparison illustrates the adverse effect of vegetation ground truth on model training.

TABLE IX

HEIGHT RANGE AND CORRESPONDING PIXEL-COUNT RANGE FOR DIFFERENT SAR IMAGES

Item	NYC_1	Wash_1	Nor_1	Nor_2
Height Range (m)	0–468	0–85	0–90	0–90
Pixel-count Range (pixels)	0–633	**0–125**	**0–110**	**0–142**

Note: Height range denotes the interval from 0 to the upper bound covering the majority of buildings. Bold values indicate smaller upper bounds of pixel-count ranges than that of *NYC_1*.

higher values. This issue is further reflected in the scatter plot, which shows a systematic underestimation at larger values.

V. DISCUSSION

A. Selection of Single-Image Training Data

In this study, the proposed model is trained using a single SAR image. The selected training region (*NYC_1*) covers a wide range of building heights and complex urban structures, providing diverse variations for model learning. To further investigate the feasibility of single-image-based estimation under simpler urban morphology, additional experiments are conducted using SAR images from Washington and Norfolk. As summarized in Table IX, *NYC_1* exhibits a significantly

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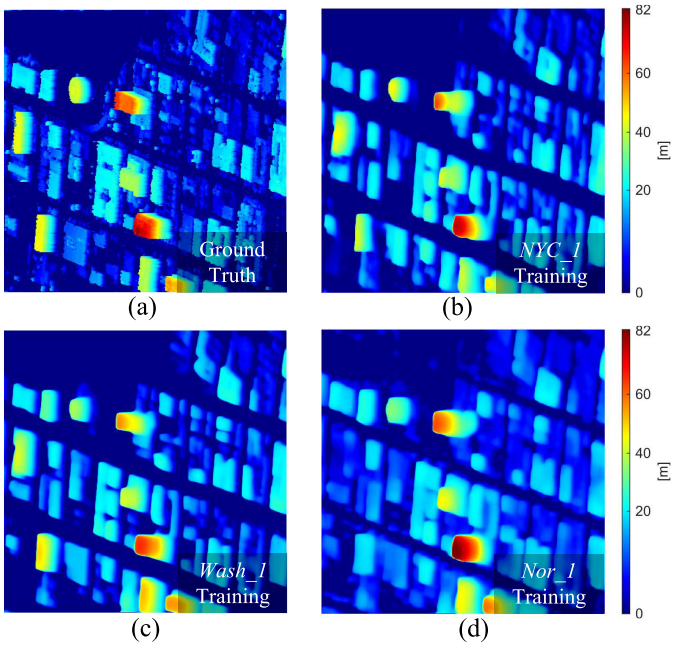


Fig. 21. Qualitative comparison on the unseen *San_1* image under single-image training from different cities. Results are shown on a 512 × 512 patch. (a) Ground truth, (b) trained on *NYC_1*, (c) trained on *Wash_1*, and (d) trained on *Nor_1*.

TABLE X
MAE RESULTS OF SINGLE-IMAGE TRAINING WITH DIFFERENT URBAN MORPHOLOGIES (M)

Test Image \ Train Image	NYC_1	Wash_1	Nor_1	Nor_2
San_1	**5.03**	5.30	6.32	6.98
San_2	**5.77**	5.97	7.23	6.99
San_3	**5.78**	6.14	7.41	7.34
San_4	7.76	**7.60**	8.26	8.79
Average	**6.09**	6.25	7.31	7.53

Note: All results are evaluated on unseen test regions, with evaluation restricted to building heights below 100 m. Bold indicates the best performance, while underlining denotes the second-best performance.

larger effective height range and pixel-count range than *Wash_1*, *Nor_1*, and *Nor_2*. As the pixel-count ranges of *Wash_1* and *Nor_1* are more restricted, twofold oversampling is applied along the range direction to extend the supported pixel-count range. The pixel-count normalization range is adjusted for each image.

The performance is evaluated on SAR images from the unseen region of San Diego, USA, with errors computed only for building heights below 100 m. The results are reported in Table X. Comparable performance can still be achieved when training on data with simpler urban morphology, although a slight degradation in accuracy is observed. This indicates that single-image-based estimation remains feasible under such conditions. However, its performance is still affected by the simpler training data distribution and the more limited height representation range of the training image. Fig. 21 presents a qualitative comparison on a 512 × 512 patch from the *San_1* image. Most buildings are still estimated correctly, with slight differences under different training settings, likely due to variations in the training data distributions. Overall, among the training settings considered in this study, the initial configuration based on *NYC_1* achieves the best performance.

B. Discussion on Cross-Sensor Generalization

Cross-sensor experiments are inherently challenging. GF-3 operates in the C-band, whereas TerraSAR-X operates in the X-band, implying different scattering characteristics. Under this setting, the proposed framework was still able to produce meaningful height estimates. By contrast, the UNet- and Transformer-based baselines did not yield sufficiently meaningful results and are therefore not shown. The cross-sensor experiments mainly serve to indicate that the proposed framework has a certain degree of generalization ability across sensors. This result, however, should not be interpreted as evidence of universal applicability across arbitrary SAR sensors. In practical applications, single-image training with data from the target sensor is still preferred.

C. Limitations of Ground Truth

A systematic error is introduced by the temporal inconsistency between the LiDAR and SAR acquisitions. Some buildings visible in the SAR images may not exist in the LiDAR data, leading to ground truth mismatches. Conversely, in some cases, LiDAR provides valid height measurements, while the corresponding buildings in the SAR image become barely visible due to their surface characteristics (e.g., highly smooth materials causing very low backscattering), appearing only as faint roof responses.

VI. CONCLUSION

In this study, we propose a tiny-dataset scale-consistent diffusion framework for single SAR image building height estimation. The framework provides an effective strategy for addressing tasks with high training costs by leveraging existing prior knowledge for task transfer. A stable diffusion-based model is fine-tuned E2E using only a single GF-3 SL SAR image as the training dataset. Besides, a scale-consistency module is designed to enable cross-image scale training and testing. In addition, the ground truth is constructed through a designed efficient pipeline, which enables the recovery of complete radar-domain building facades from sparse LiDAR point clouds. Furthermore, a data augmentation strategy improves performance under tiny-dataset training. Experiments confirm that our framework achieves optimal estimation under limited training data, outperforming baselines even when they are trained on larger datasets. Extended experiments on TerraSAR-X further demonstrate the cross-sensor generalization capability of the proposed framework.

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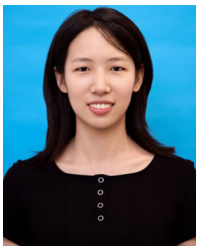
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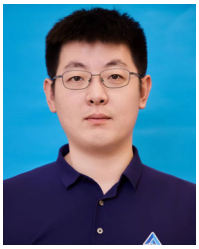
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